



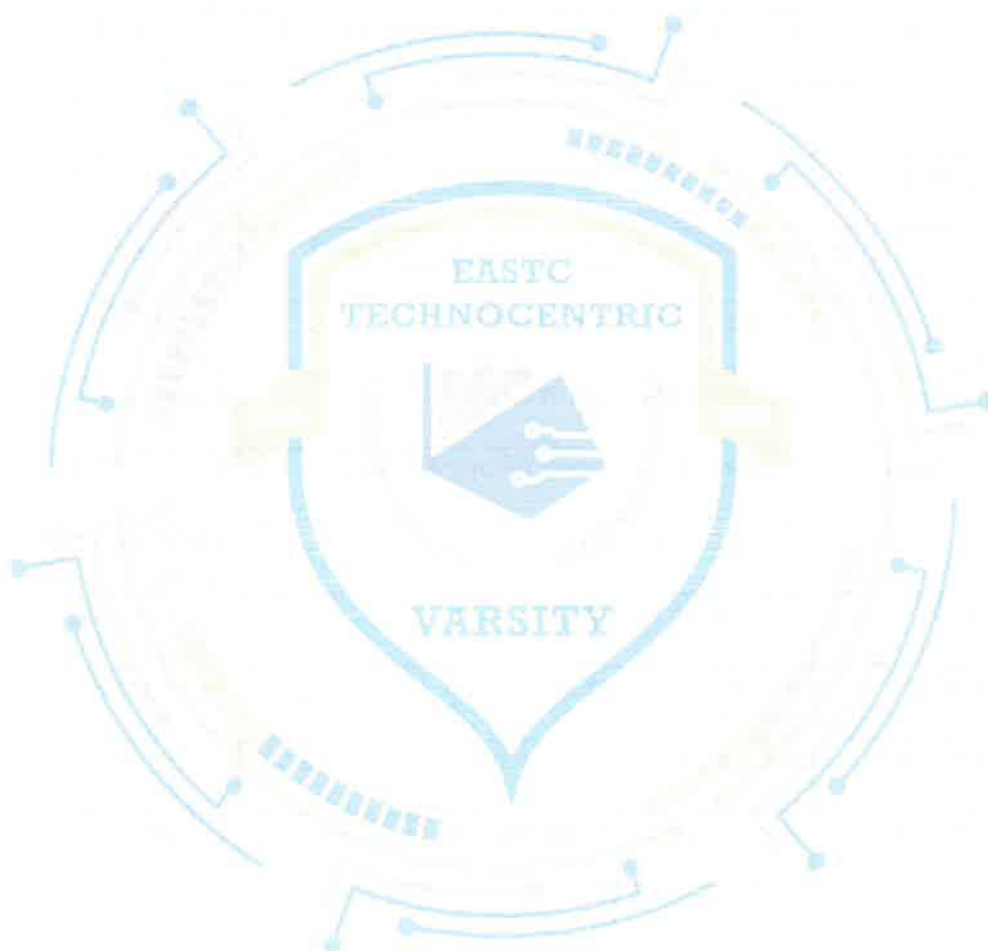
ELECTRICITY AND ELECTRONICS
SUMMATIVE ASSESMENTS

Competent
 $\frac{56}{60}$
93%

Module #	671101000-KM-03:
NQF Level	level 4,
Notional hours	130
Credit(s)	Cr 13
Occupation or Specialisation	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

Welcome to your Summative Assessment!

This document deals with the summative assessment of your competence towards the outcomes which you need to achieve. "Summative Assessment is assessment for making a judgement about achievement. This is carried out when a learner is ready to be assessed at the end of a programme of learning" (SAQA: Criteria and Guidelines for Assessment Policy Document,).



Special arrangements for assessmentPlace EastcLanguage English

Resources

Barriers NA**People to be involved with assessment**Learner: Lerato

Manager:

Trainer: R Nyakala

Mentor / Coach:

Assessor:

Moderator:

Next steps for learning

Resources required for this assessment

Guidelines to the learner:

Learners Name: Lerato m DikshicoLearner's signature: Dikshico Date: _____

Assessors Name: _____

Assessor's signature: _____ Date: _____

ASSESSMENT APPEAL APPLICATION FORM

LEARNER'S NOTICE OF ASSESSMENT APPEAL

TO: The Internal Moderator

A meeting with the internal moderator is hereby requested to discuss the outcome of my assessment.

Internal moderator name _____

Date of submission: _____

Name of employee assessed: _____

Name of Assessor: _____

Date of feedback session: _____

Grounds for Appeal

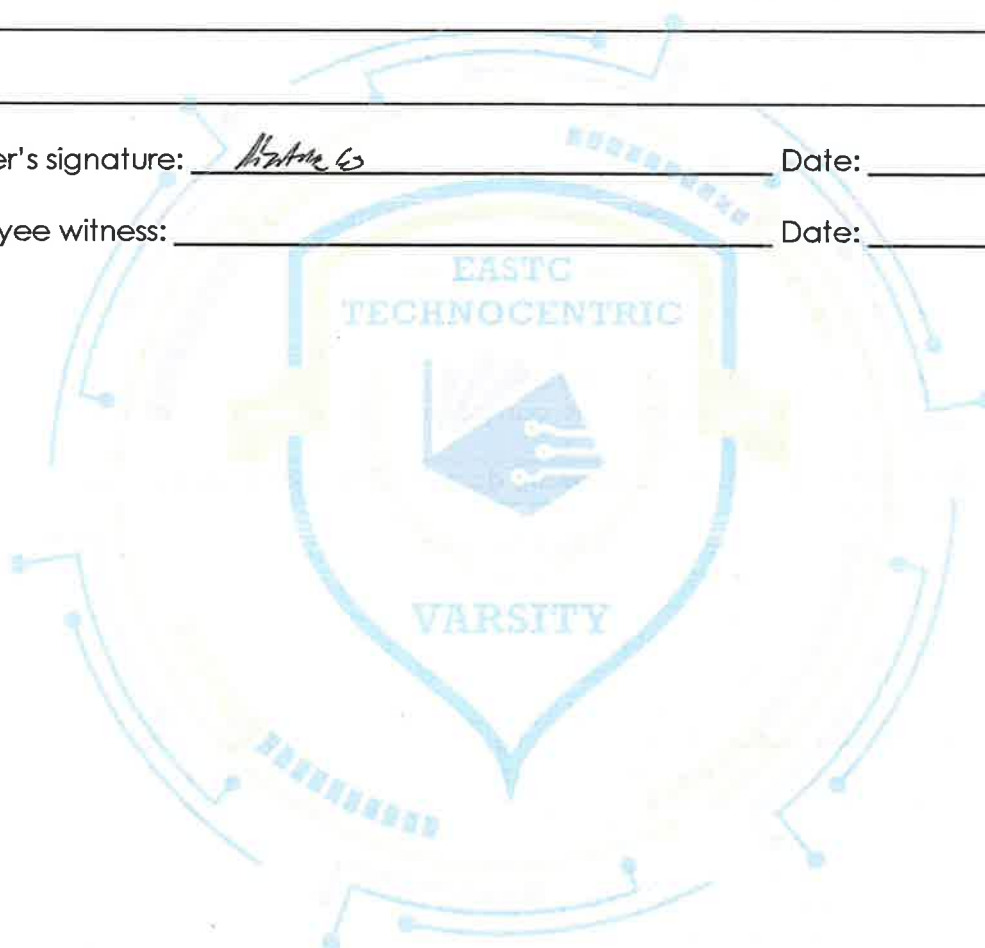
No	Tick the applicable ground(s) for appeal	Tick
1	The assessment did not follow the laid down procedure	
2	Not all evidence available was taken into account during the assessment	
3	The assessor was not a subject matter expert nor was a subject matter expert present during the assessment process	
4	The assessment was not according to the performance criteria and the range statement stipulated in the unit standard	
5	Not all the range items were available for the assessment	

Reasons for Appeal

No	Please give detailed reasons for the choice(s) above
1	
2	
3	
4	
5	

Learner's signature: *M. A. M. G.* Date: _____

Employee witness: _____ Date: _____



Learner declaration of acceptance of assessment instruments and relevant documentation: Date:

Learners Name:

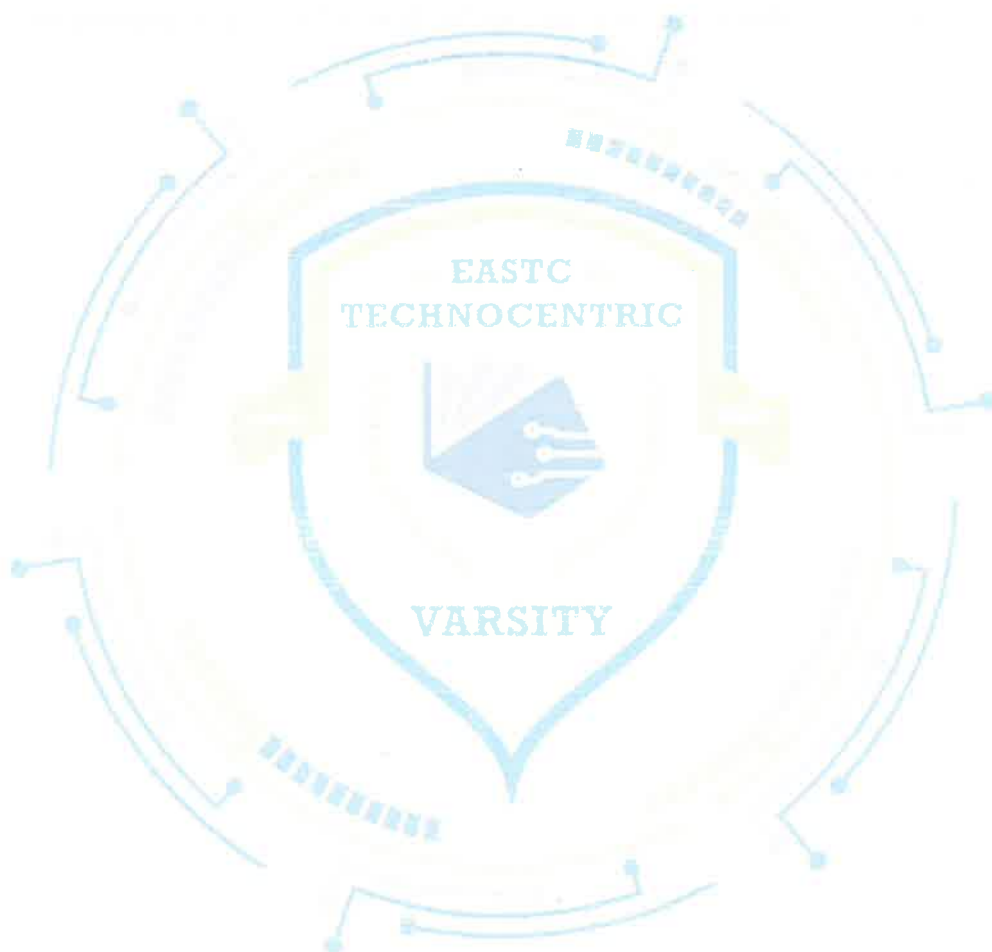
Leano

Signature

Mphahlele

Assessors Name:

Signature:



LEARNING UNIT 1:

KM-03-KT01: Fundamentals of electricity (35%)

Learning Outcome 1:	Principles and fundamental concepts of electricity
Learning Outcome 2:	Definitions, types, properties and applications of conductors, insulators and semi-conductors
Learning Outcome 3:	Concepts, theories and principles of Electrical Circuits
Learning Outcome 4:	Calculations on basic electrical circuits (resistance, voltage and current) and power
Learning Outcome 5:	Basic principles and calculation of magnetism

Task	Questions Description	Marks
1	List types of materials used for conductors, insulators and semi-conductors and describe their mechanical and electrical properties and applications.	10

1 electrical conduct

copper

silver Gold Brass

Aluminium

2 electrical properties

Very low electrical resistance

Allow electric current to flow easily

high electrical conductivity

mechanical properties

3 Duralite (can be shaped)

Strong and durable

Good heat conductor

Applications

wire cable

4 Insulation

material used

Task	Questions Description	Marks
2	a) Name, describe and explain the different types of magnets and their properties b) Explain fundamental magnetic concepts by naming the five characteristics of magnetic lines of force and explaining the relationship between flux and flux density	10

1- Permanent magnets : these retain their magnetism for long periods without an external power source e.g (Alnico)

2- Temporary magnets these act like magnets only while within a magnetic field.

3- Electromagnets : these consist of a wire coil wrapped around a metal core they are only magnetic when an electric current flows through the coil their strength

& can be easily controlled by changing the current

b) Fundamental magnetic concepts

they always form continuous closed loops

they never cross or intersect each other

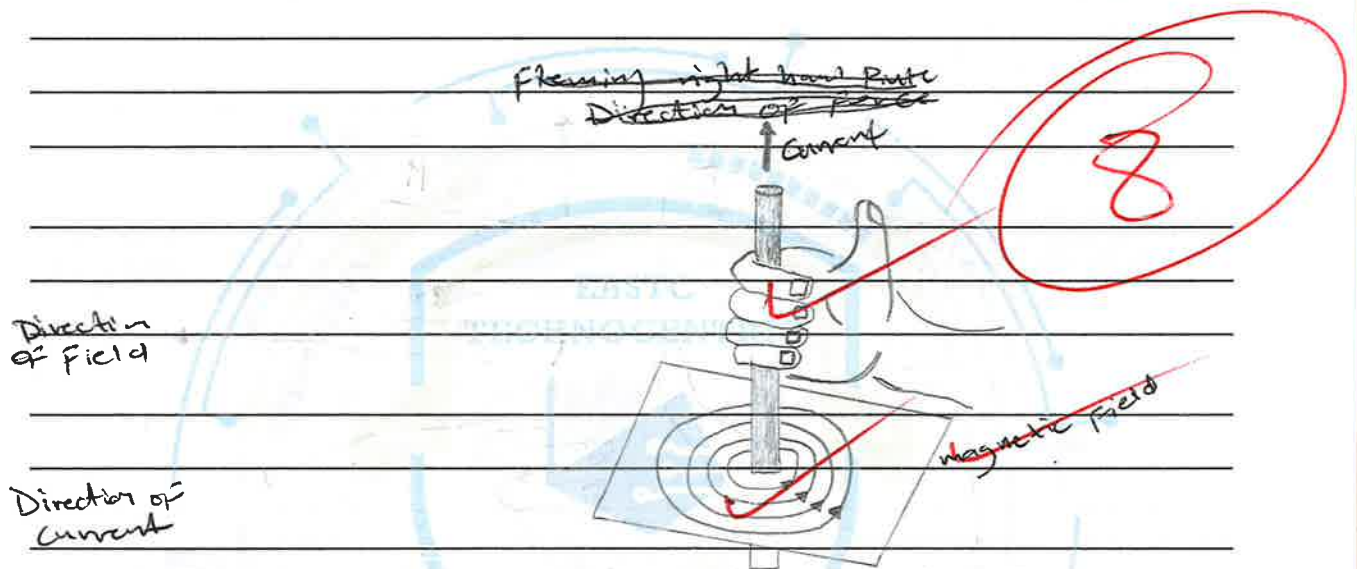
they travel from the north pole to the south pole

Outside the magnetic

they seek the path of least resistance reluctance

they act like stretched rubber bands trying to short themselves

Task	Questions Description	Marks
3	Describe, with the aid of a drawing, the application of the right-hand grip rule and show how a magnetic field is established when an electrical current flows through a conductor, by using formula, calculate the force on a current carrying conductor.	10



the right-hand Grip rule (also known as the rule) is a simple mnemonic used to determine the direction of the magnetic field around a straight current-carrying wire

Thumb: point your right thumb in the direction of the conventional current flow positive to negative

The fingers: curls your finger around the imaginary conductor

The field: curl your finger in which your fingers curl indicates the direction in which your finger curl indicate the direction of the magnetic flux line (the magnetic field)

establishing a magnetic field when DC current flows through a conductor it doesn't just stay inside the wire it

Creates a magnetic field in the surround space. concentric circle

$F = B I L \sin(\theta)$ F = Force acting on the conductor
 B magnetic flux I = current L = length

- Types of earthing systems

e.g. (TNs, TN-C-S, TT) these systems define how the neutral and earthing conductors are connected

- Earthing electrode requirement

size, type and installation methods for grounding electrodes (rods, plates etc)

- Bonding requirements

bonding conductive parts of electrical

installations, metallic pipes and other conductive structure

- Testing and inspection

to ensure the effectiveness of earthing and bonding system

Protective devices

requirement for use of protective devices like circuit breakers and residual current devices (RCD) to further enhance safety

LEARNING UNIT 3:

KM-03-KT02: Electronics (30%)

Learning Outcome 1:	International Electro technical Commission (IEC) symbols for electronic components
Learning Outcome 2:	Principles, safety precautions, identification and basic function of electronic components
Learning Outcome 3:	Basic electronic circuits
Learning Outcome 4:	Principles of rectification, single phase and 3-phase

Task	Questions Description	Marks
6	a) Describe the various inductors used and typical applications b) Describe the construction and uses of various semiconductor devices c) Draw basic electronic circuits using International Electrotechnical Commission (IEC) symbols	10 9

Inductors store energy in a magnetic field when electric current flows through them. Their performance depends largely on the core material used.

Inductor type	Description	Typical Application
Air core	Coils wound on non-magnetic	high-frequency Application like RF tuning and radio transmission

B Construction and uses of semiconductor devices

- Semiconductors are the building blocks of modern electronics
- Usually made from silicon or germanium doped with impurities
- Diode - construction created by joining a P-type material (excess-holes) and an N-type material joining P-type material P-N junction

C. i. Simple RC Circuit



Resistor



Inductor



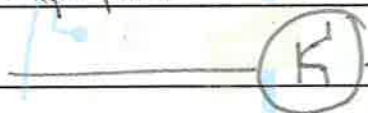
Capacitor

2 Rectifier Circuit



Diode

3 Amplifier



Transistor

Assessor's Report

Candidate's Name	Lerato	ID No.	880916 0416 083
Assessor's Name		Reg. No.	
Title	Electricity and electronic		
ASSESSMENT DECISION			
Outcome	C	NYC	Comments
OVERALL COMPETENCE JUDGEMENT:			
Comments			
Date			
	L. Mhoneko		
Signature of Assessor	Signature of Candidate		

Candidate Appeal Form

Candidate's Name:		ID No.	
Assessor's Name:		Reg. No.	
Title:			
Date:			
SECTION 1			
Candidate's reason for disagreeing with the assessment decision			
Assessor's rationale for the assessment decision			
Candidate's signature			
Assessor's signature			
SECTION 2			
Moderator's decision			
Moderator's signature			

ASSESSMENT REVIEW

NAME of LEARNER <i>Lerato</i>		NAME of ASSESSOR	
VENUE <i>Eastc</i>		DATE of REVIEW	
Review Dimension	ASSESSOR	LEARNER/ CANDIDATE	ACTION
The principles/criteria for good assessment were achieved?	☐ Agree ☐ Disagree	☐ Agree ☐ Disagree	
The assessment related to the registered unit standard?	☐ Agree ☐ Disagree	☐ Agree ☐ Disagree	
The assessment was practical?	☐ Agree ☐ Disagree	☐ Agree ☐ Disagree	
It was time efficient and cost-effective and did not interfere with my normal responsibilities?	☐ Agree ☐ Disagree	☐ Agree ☐ Disagree	
The assessment instruments were fair, clear and understandable	☐ Agree ☐ Disagree	☐ Agree ☐ Disagree	
The assessment judgements was made against set requirements	☐ Agree ☐ Disagree	☐ Agree ☐ Disagree	
The venue and equipment was functional?	☐ Agree ☐ Disagree	☐ Agree ☐ Disagree	
Special needs were identified and the assessment plan was adjusted	☐ Agree ☐ Disagree	☐ Agree ☐ Disagree	
Feedback was constructive against the evidence required	☐ Agree ☐ Disagree	☐ Agree ☐ Disagree	
An opportunity to appeal was given	☐ Agree ☐ Disagree	☐ Agree ☐ Disagree	
The evidence was recorded	☐ Agree ☐ Disagree	☐ Agree ☐ Disagree	
LEARNER'S DECLARATION OF UNDERSTANDING			
I am aware of the moderation process and understand that the moderator could declare the assessment decision invalid			
<i>Lerato</i>			
Learner	Date	Assessor	Date
Moderator		Date	